

ESE PYQ CC Sem 6

Jan 25 - Apr 25

| Que. No. | Question                                                                             | Max. Marks |
|----------|--------------------------------------------------------------------------------------|------------|
| Q1       | Solve any <b>Four</b>                                                                | <b>20</b>  |
| i)       | Define cloud computing. List any four essential characteristics.                     | 5          |
| ii)      | What are the main contributions of Salesforce.com to cloud-based business solutions? | 5          |
| iii)     | Compare and contrast Xen, VMware in terms of architecture and performance.           | 5          |
| iv)      | Explain different types of cloud deployment models with suitable examples.           | 5          |
| v)       | Write a short note on Google App Engine's sandbox environment.                       | 5          |
| vi)      | List and explain two productivity applications that use the cloud.                   | 5          |

| Que. No. | Question                                                                                                     | Max. Marks |
|----------|--------------------------------------------------------------------------------------------------------------|------------|
| Q2 A     | Solve the following                                                                                          | <b>10</b>  |
| i)       | Describe DevOps practices in cloud-based application development and deployment.                             | 5          |
| ii)      | Explain any two economic benefits of cloud computing.                                                        | 5          |
|          | OR                                                                                                           |            |
| Q2 A     | Discuss the structured and non-structured storage solutions provided by Amazon AWS.                          | 10         |
| Q 2 B    | Solve any <b>One</b>                                                                                         | <b>10</b>  |
| i)       | What is a virtualization? Further, State its importance in Cloud Computing. Explain types of Virtualization. | 10         |
| ii)      | Explain the Cloud Reference Model and the different types of service models.                                 | 10         |

| Que. No. | Question                                                                      | Max. Marks |
|----------|-------------------------------------------------------------------------------|------------|
| Q3       | Solve any <b>Two</b>                                                          | <b>20</b>  |
| i)       | Explain various compute, storage, and communication services provided by AWS. | 10         |
| ii)      | What are the key features of Google Cloud Platform?                           | 10         |
| iii)     | Discuss cloud-based media and gaming applications.                            | 10         |

| Que. No. | Question                                           | Max. Marks |
|----------|----------------------------------------------------|------------|
| Q4       | Solve any <b>Two</b>                               | <b>20</b>  |
| i)       | How is cloud used for ECG analysis in healthcare?  | 10         |
| ii)      | Define microservices and explain their advantages. | 10         |
| iii)     | Describe Dockers and Container.                    | 10         |

| Que. No. | Question              | Max. Marks |
|----------|-----------------------|------------|
| Q5       | Short Note (any Four) | <b>20</b>  |
| i)       | CRM                   | 5          |
| ii)      | ERP                   | 5          |
| iii)     | SAAS                  | 5          |
| iv)      | Virtualization        | 5          |
| v)       | MetaCDN               | 5          |
| vi)      | Federated Clouds      | 5          |

Jan 24 - Apr 24

| Que. No. | Question                                                                         | Max. Marks |
|----------|----------------------------------------------------------------------------------|------------|
| Q1       | Solve any <b>Four</b>                                                            | <b>20</b>  |
| i)       | What are the key characteristics of cloud computing?                             | 5          |
| ii)      | Describe the Infrastructure as a Service (IaaS) model.                           | 5          |
| iii)     | Differentiate Public cloud and Private cloud.                                    | 5          |
| iv)      | Discuss third party cloud Services.                                              | 5          |
| v)       | Explain Google Cloud Platform.                                                   | 5          |
| vi)      | Describe NIST (National Institute of Standards and Technology) cloud definition. | 5          |

| Que. No. | Question                                                                                      | Max. Marks |
|----------|-----------------------------------------------------------------------------------------------|------------|
| Q2 A     | Solve the following                                                                           | <b>10</b>  |
| i)       | Explain Cloud Reference model.                                                                | 5          |
| ii)      | Write a note on Manjrasoft Aneka.                                                             | 5          |
|          | OR                                                                                            |            |
| Q2 A     | Compare the compute, storage, and communication services of AWS and Google Cloud Platform?    | 10         |
| Q 2 B    | Solve any <b>One</b>                                                                          | <b>10</b>  |
| i)       | Describe the role of virtualization in cloud computing?                                       | 10         |
| ii)      | Discuss some popular cloud-computing providers, and what distinguishes them from one another? | 10         |

| Que. No. | Question                                                                                                       | Max. Marks |
|----------|----------------------------------------------------------------------------------------------------------------|------------|
| Q3       | Solve any <b>Two</b>                                                                                           | <b>20</b>  |
| i)       | Explain the need of Virtualization in the context of Cloud Computing. Compare type-1 and type-2 Virtualization | 10         |
| ii)      | Explain the concepts of Dockers, containers, and microservices in cloud computing?                             | 10         |
| iii)     | Discuss the key features and benefits of Microsoft Hyper-V in the context of virtualization?                   | 10         |



| Que. No. | Question                                                                                                       | Max. Marks |
|----------|----------------------------------------------------------------------------------------------------------------|------------|
| Q4       | Solve any <b>Two</b>                                                                                           | <b>20</b>  |
| i)       | Explain in brief about Amazon web services (AWS).                                                              | 10         |
| ii)      | Illustrate the concept of scientific application using cloud for ECG analysis.                                 | 10         |
| iii)     | Discuss some challenges and considerations when deploying multiplayer online gaming applications in the cloud? | 10         |

| Que. No. | Question                       | Max. Marks |
|----------|--------------------------------|------------|
| Q5       | Write notes on any <b>four</b> | <b>20</b>  |
| i)       | Hadoop                         | 5          |
| ii)      | SalesForce.com                 | 5          |
| iii)     | Google AppEngine               | 5          |
| iv)      | ERP                            | 5          |
| v)       | SpotCloud                      | 5          |
| vi)      | DevOps                         | 5          |



Jan 24 - Apr 24 Section B

| Que. No. | Question                                  | Max. Marks |
|----------|-------------------------------------------|------------|
| Q 4      | Solve any <b>Four</b>                     | 20         |
| i)       | Discuss Amazon Web Services (AWS).        | 5          |
| ii)      | Explain Google Cloud platform (GCP).      | 5          |
| iii)     | Explain Google App Engine (GAE).          | 5          |
| iv)      | Explain the concept of Microsoft Hyper-V. | 5          |
| v)       | Explain Cloud Reference Model.            | 5          |
| vi)      |                                           | 5          |

| Que. No. | Question                       | Max. Marks |
|----------|--------------------------------|------------|
| Q 5      | Write notes on any <b>four</b> | 20         |
| i)       | ERP.                           | 5          |
| ii)      | SpotCloud.                     | 5          |
| iii)     | DevOps.                        | 5          |
| iv)      | ECG using Cloud Computing.     | 5          |
| v)       | Multiplayer Online Gaming.     | 5          |
| vi)      | Protein Structure Prediction.  | 5          |

| Que. No. | Question                                                                                                                                                                                      | Max. Marks |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| Q 6      | Objective/Multiple Choice Questions                                                                                                                                                           | 10         |
| 1        | Which Google Cloud service is designed for real-time messaging and event ingestion?<br>a) Google Cloud Pub/Sub<br>b) Google Cloud Storage<br>c) Google Cloud Datastore<br>d) Google Cloud SQL |            |
| 2        | What is Cloud Dataflow used for in Google Cloud Platform?<br>a) Real-time data analysis and processing<br>b) Virtual machine management<br>c) Container orchestration<br>d) File storage      |            |
| 3        | What is a hypervisor?<br>a) A software that manages virtual machines and their resources<br>b) A physical server<br>c) A type of virtual machine                                              |            |

|    |                                                                                                                                                                                                                                                                                                                                      |  |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|    | d) A network protocol                                                                                                                                                                                                                                                                                                                |  |
| 4  | <p>What is virtualization?</p> <p>a) A technology that creates multiple virtual instances of physical hardware</p> <p>b) A technique used to simulate physical environments in software</p> <p>c) The process of converting physical servers into virtual machines</p> <p>d) All of the above</p>                                    |  |
| 5  | <p>Which of the following is a benefit of server virtualization?</p> <p>a) Increased hardware costs</p> <p>b) Decreased server utilization</p> <p>c) Improved resource utilization</p> <p>d) Limited scalability</p>                                                                                                                 |  |
| 6  | <p>What is live migration in virtualization?</p> <p>a) Moving a virtual machine from one physical server to another without disrupting service</p> <p>b) Shutting down a virtual machine for maintenance</p> <p>c) Creating a new virtual machine from a template</p> <p>d) Allocating additional resources to a virtual machine</p> |  |
| 7  | <p>What is Amazon S3?</p> <p>a) A relational database service</p> <p>b) A serverless computing platform</p> <p>c) An object storage service</p> <p>d) A content delivery network</p>                                                                                                                                                 |  |
| 8  | <p>What is Amazon EC2?</p> <p>a) An object storage service</p> <p>b) A serverless computing service</p> <p>c) A virtual machine service</p> <p>d) A content delivery network</p>                                                                                                                                                     |  |
| 9  | <p>What is Amazon RDS?</p> <p>a) A NoSQL database service</p> <p>b) A data warehousing service</p> <p>c) A managed relational database service</p> <p>d) A serverless computing service</p>                                                                                                                                          |  |
| 10 | <p>Which virtualization technology allows multiple virtual servers to share physical hardware resources while maintaining isolation?</p> <p>a) Network virtualization</p> <p>b) Storage virtualization</p> <p>c) Server virtualization</p> <p>d) Desktop virtualization</p>                                                          |  |

Jan 23 - May 23

| Que. No. | Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Max. Marks |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| Q1       | Solve any <b>Four</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>20</b>  |
| i)       | Define Cloud Computing. Explain its four deployment models.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 5          |
| ii)      | Explain the concept of sandboxing in Google AppEngine. Give the examples of some of the operations not allowed in the sandbox.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 5          |
| iii)     | State IaaS, PaaS and SaaS with respect to what they provide to the customer, product types associated with them and Vendors with their specific products providing them.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 5          |
| iv)      | Explain how cloud computing technology is useful in productivity related applications giving examples.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 5          |
| v)       | Explain the concept of containerization. What are its advantages over virtualization?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 5          |
| vi)      | An eCommerce company needs to build its own IT infrastructure so as to facilitate scalability when needed and maximum uptime for its services. Further, it wants to provision the computing resources to all its departments like R&D, Analytics, Marketing etc. from the same infrastructure. It is desirable that the development work, experimentation going on in the different departments is properly isolated in their respective VMs and the problem in any VM doesn't affect the functioning of VMs of the other departments. Moreover, it would like the VMs to be independent of any host OS so as to make them isolated from its limitations. The company would like to buy heterogeneous types of processors so that it can cater to all kinds of computing needs. The company is also ready to recruit the experts for deployment and maintenance if all the above-mentioned requirements lead to complexity. Recommend the virtualization solution including mechanism, structure of virtualization for this company. Clearly justify your choice by stating how it addresses each specification given above. | 5          |

| Que. No. | Question                                                                                                                                  | Max. Marks |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------|------------|
| Q2 A     | Solve the following                                                                                                                       | <b>10</b>  |
| i)       | Explain the concept of Federated clouds. Explain the needs in cloud computing that leads to the concept of Federated Cloud.               | 5          |
| ii)      | What is energy efficient computing? Why is it fundamental in Cloud Computing?                                                             | 5          |
|          | OR                                                                                                                                        |            |
| Q2 A     | Explain the Market Oriented Cloud Computing model with a neat diagram.                                                                    | 10         |
| Q 2 B    | Solve any <b>One</b>                                                                                                                      | <b>10</b>  |
| i)       | Explain the following with respect to Amazon S3: a) Resource naming, b) Buckets, c) Objects and Metadata, d) Access control and security. | 10         |
| ii)      | Explain the architecture of Google AppEngine with a neat diagram.                                                                         | 10         |



| Que. No. | Question                                                                                                                         | Max. Marks |
|----------|----------------------------------------------------------------------------------------------------------------------------------|------------|
| Q3       | Solve any <b>Two</b>                                                                                                             | <b>20</b>  |
| i)       | Compare Microsoft Hyper V and VMWare vSphere.                                                                                    | 10         |
| ii)      | Explain the challenges in fully adopting cloud computing. Explain each one giving examples.                                      | 10         |
| iii)     | What issues/challenges does the logical and operational level of the Cloud Federation Stack address? Explain each one in detail. | 10         |

| Que. No. | Question                                                                                                                                                                               | Max. Marks |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| Q4       | Solve any <b>Two</b>                                                                                                                                                                   | <b>20</b>  |
| i)       | What are the third-party cloud services? Explain MetaCDN in detail.                                                                                                                    | 10         |
| ii)      | Explain the high-level architecture for supporting energy-efficient resource allocation in Green Cloud Computing infrastructure with a neat diagram. Explain each component in detail. | 10         |
| iii)     | Explain the Amazon EC2 instance life cycle.                                                                                                                                            | 10         |

| Que. No. | Question                                                                                                                                                                           | Max. Marks |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| Q5       | <b>Attempt any four of following:</b>                                                                                                                                              | <b>20</b>  |
| i)       | What do you understand by the term service with respect to cloud computing? What are the characteristics of a good service? Explain each characteristic in detail giving examples. | 5          |
| ii)      | Define virtualization. Explain in brief the desktop, network, data and application virtualizations.                                                                                | 5          |
| iii)     | Explain the application services available in Google AppEngine.                                                                                                                    | 5          |
| iv)      | Explain how cloud computing technology is useful in media applications giving examples.                                                                                            | 5          |
| v)       | What is the vendor lock-in problem? What effect does it have on the commercial aspects of Cloud Computing?                                                                         | 5          |
| vi)      | Explain the concept of Virtual Market Place. What are its main features?                                                                                                           | 5          |